

Convolutional filters

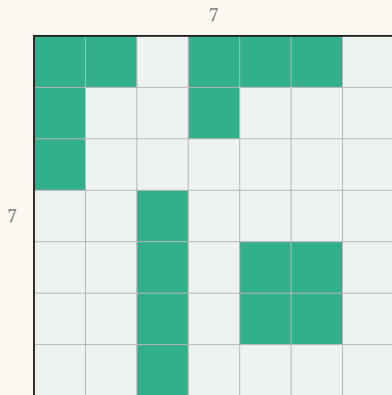
Weekend 3, Friday morning

Carlos Cotrini

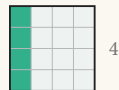
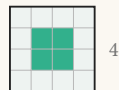
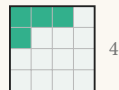
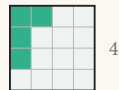
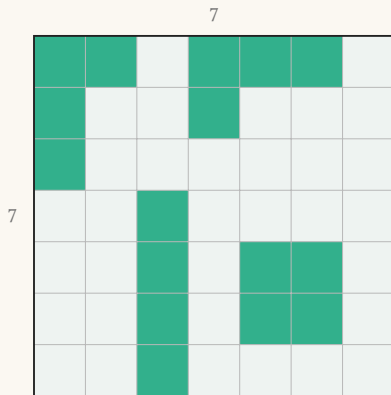
CAS Building ML/AI Applications, ETH Zurich

Friday 2 October 2026

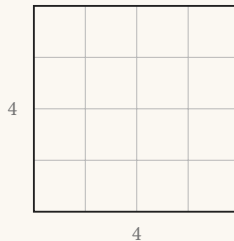
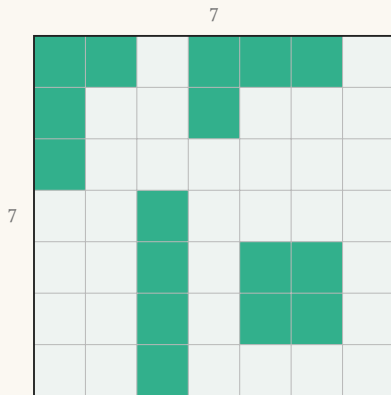
Decomposing an image into tiles



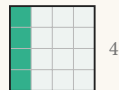
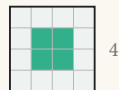
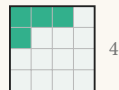
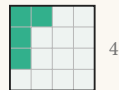
Decomposing an image into tiles



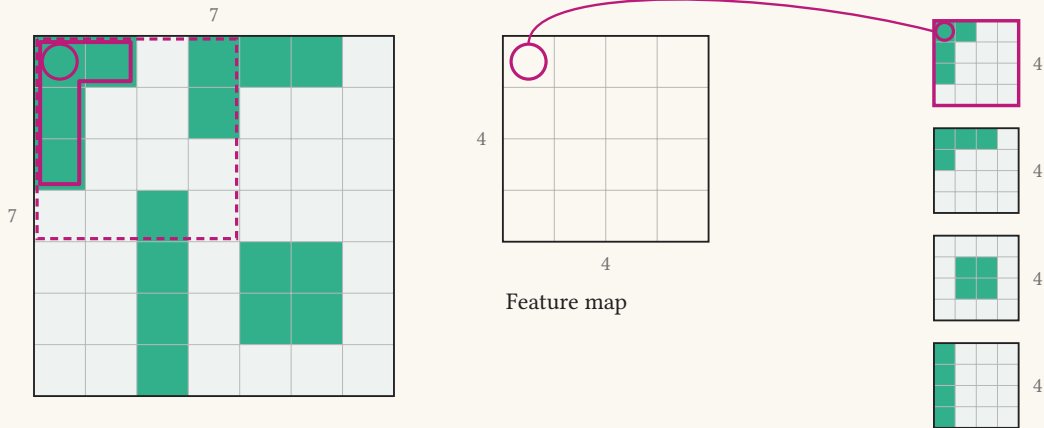
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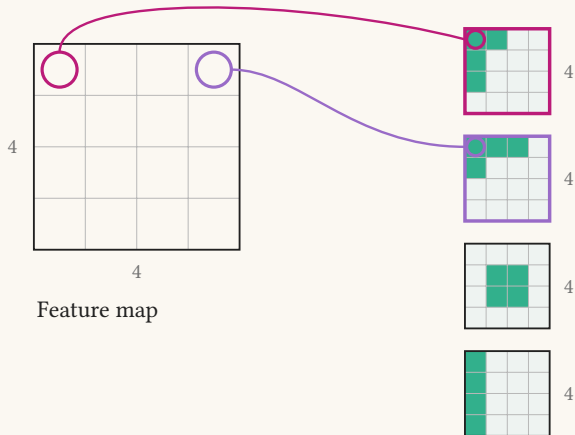
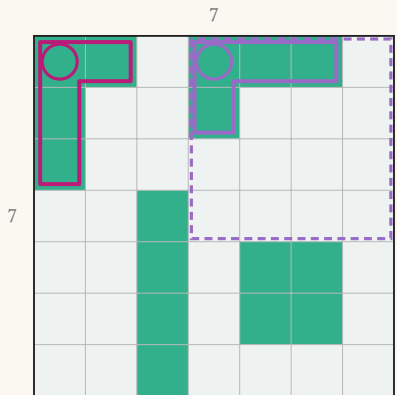
Feature map



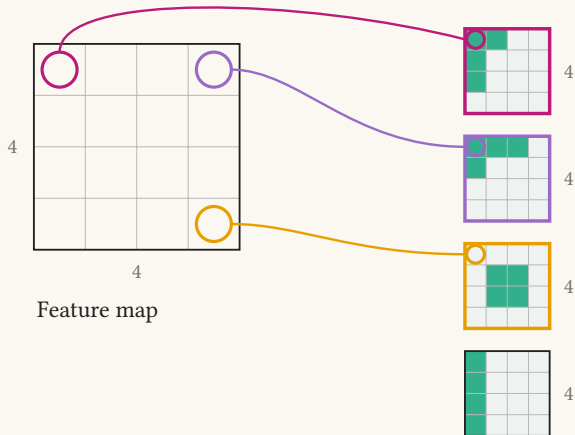
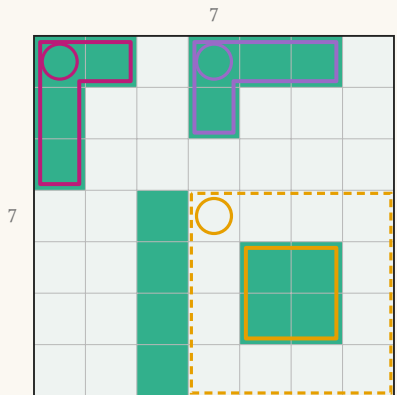
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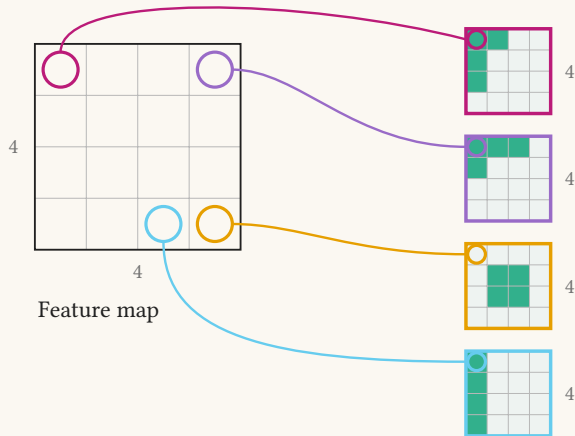
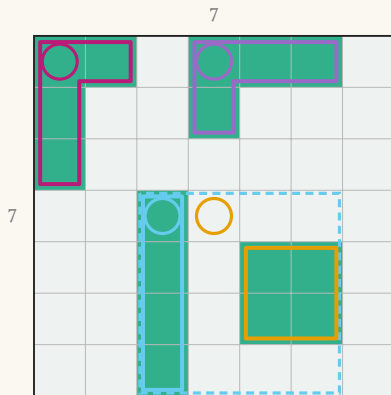
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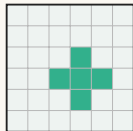
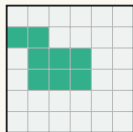
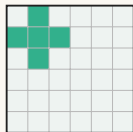
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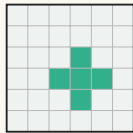
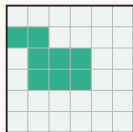
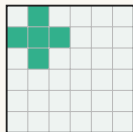
Decomposing an image into tiles



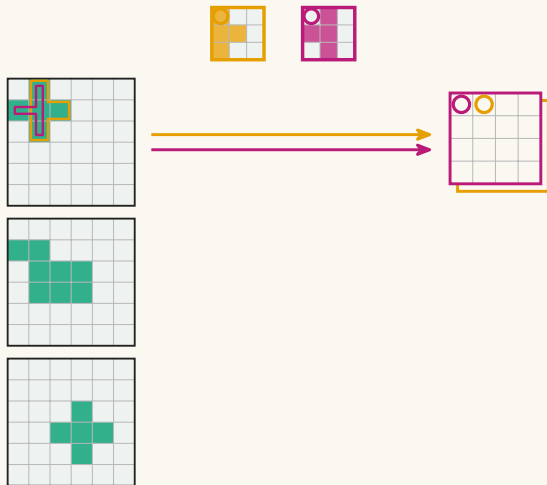
Recognizing islands that look like +



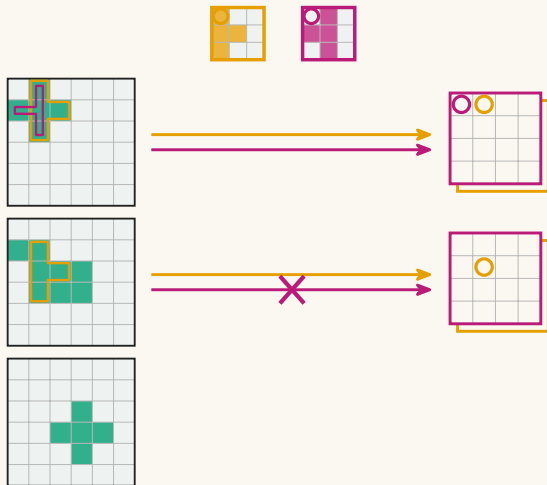
Recognizing islands that look like +



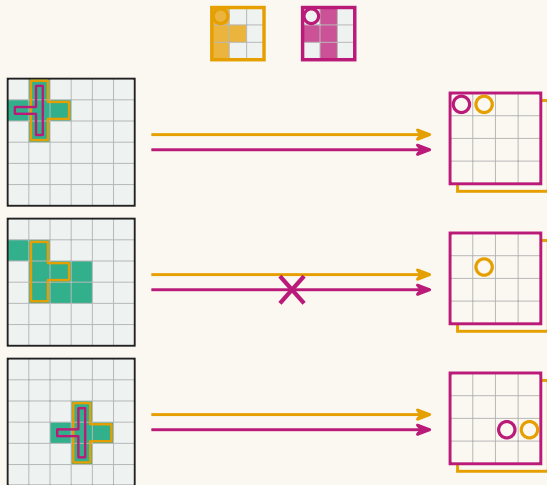
Recognizing islands that look like +



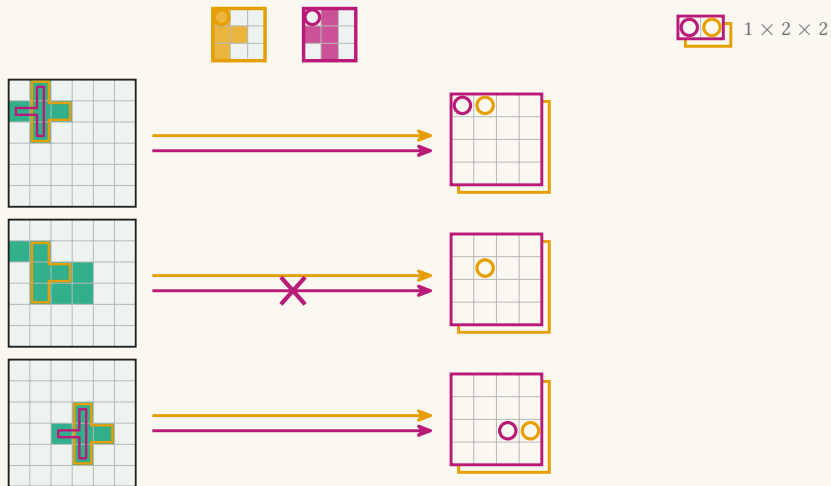
Recognizing islands that look like +



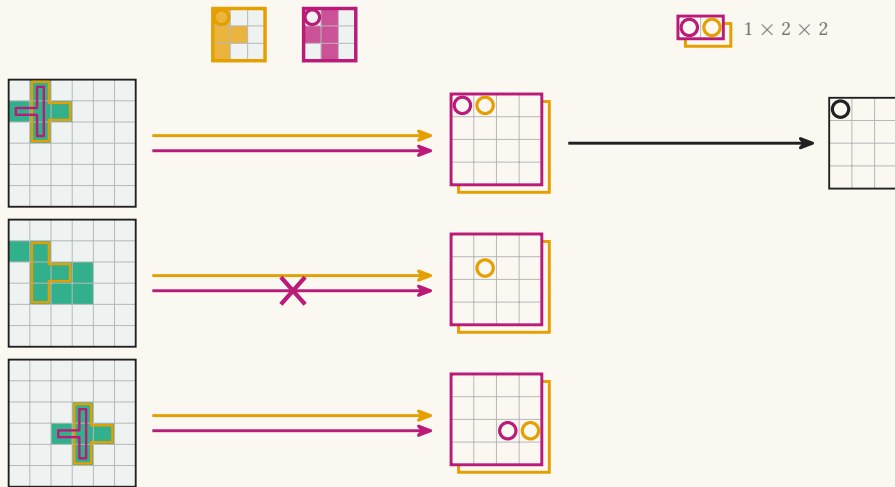
Recognizing islands that look like +



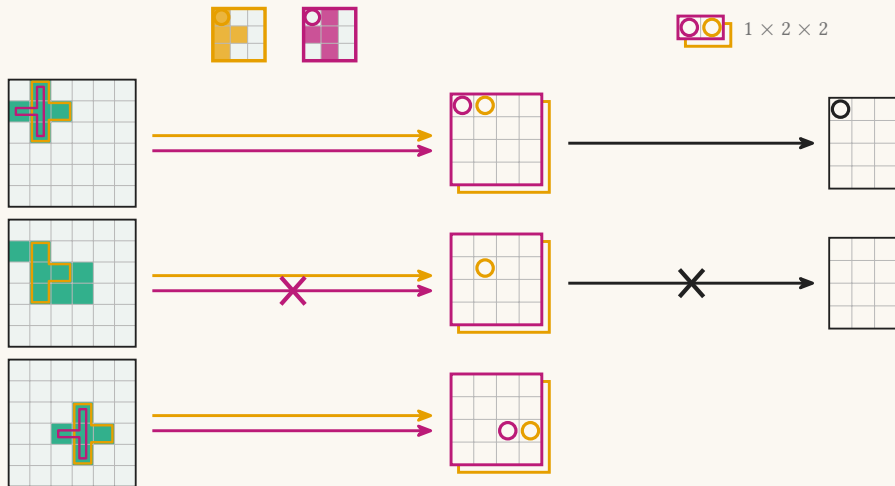
Recognizing islands that look like +



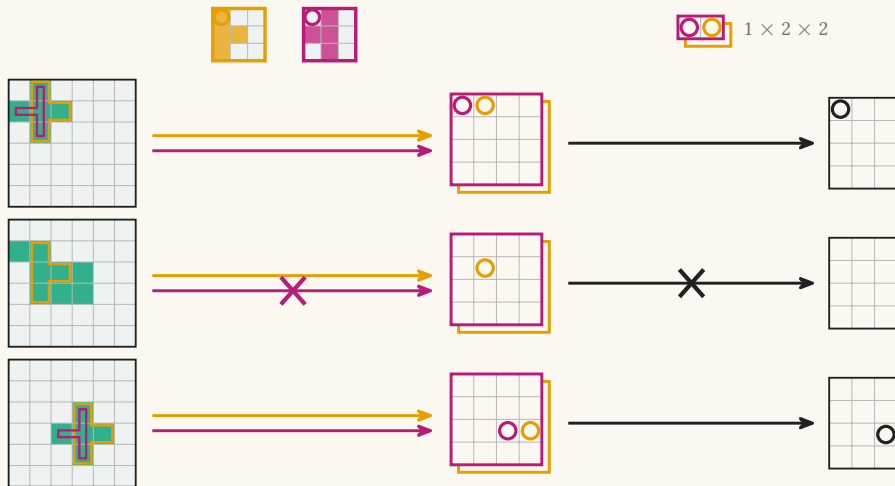
Recognizing islands that look like +



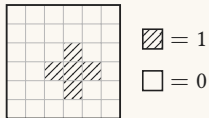
Recognizing islands that look like +



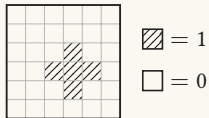
Recognizing islands that look like +



How to compute activations



How to compute activations



0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0

How to compute activations



 = 1

 = 0

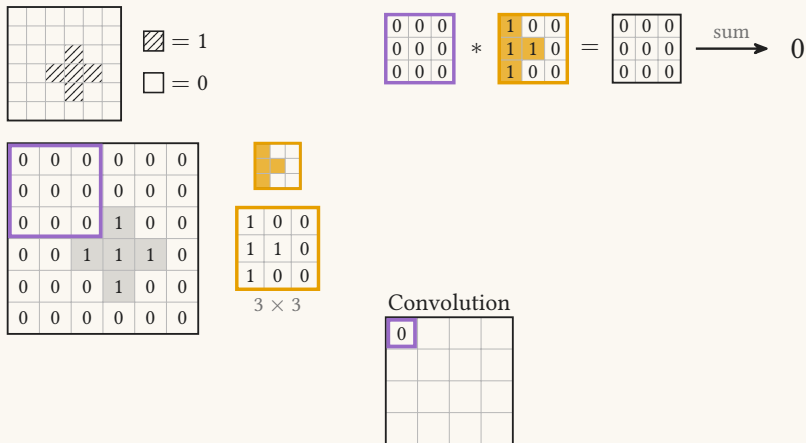
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
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0	0	0	0	0	0



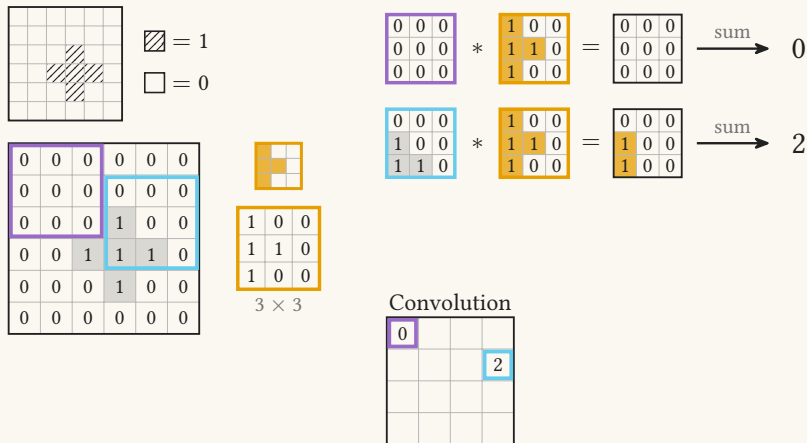
1	0	0
1	1	0
1	0	0

3×3

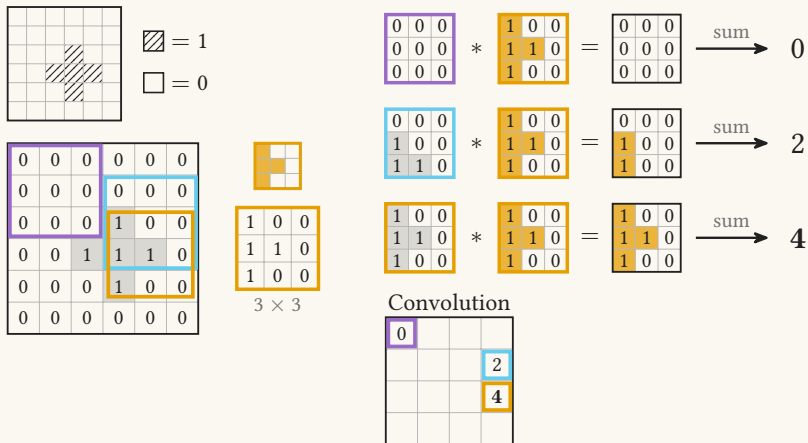
How to compute activations



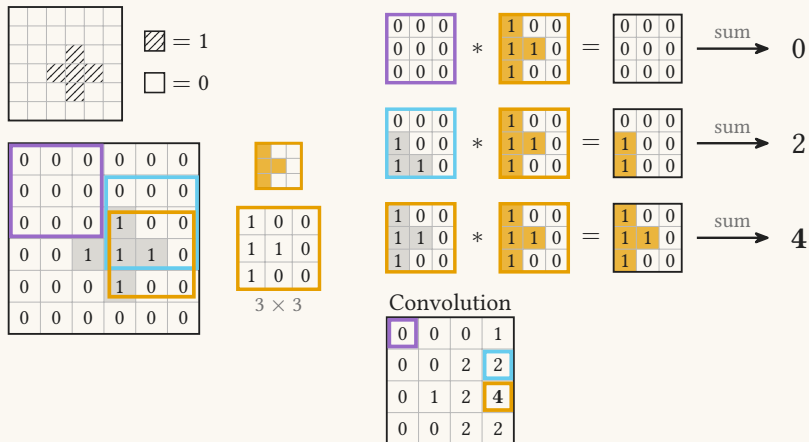
How to compute activations



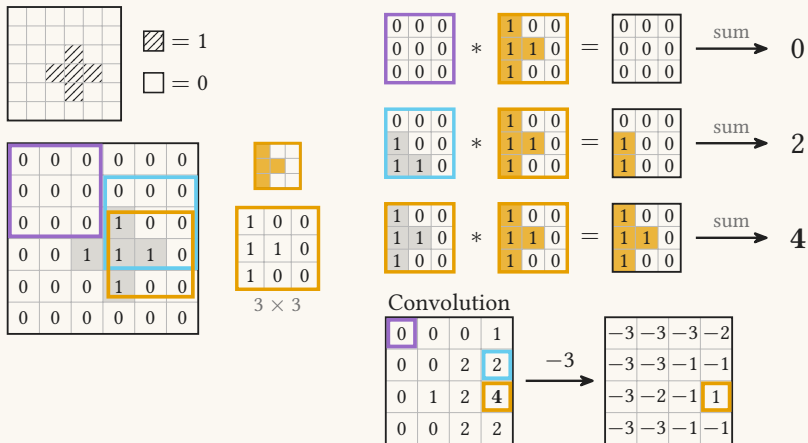
How to compute activations



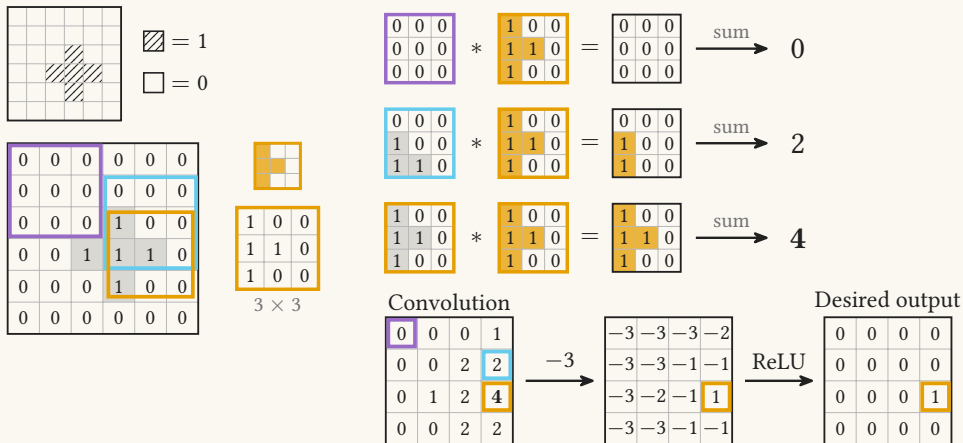
How to compute activations



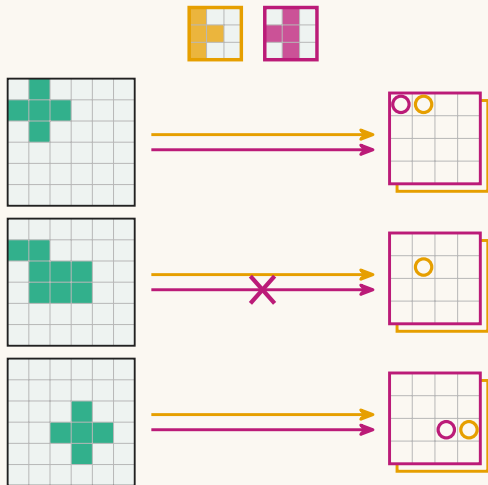
How to compute activations



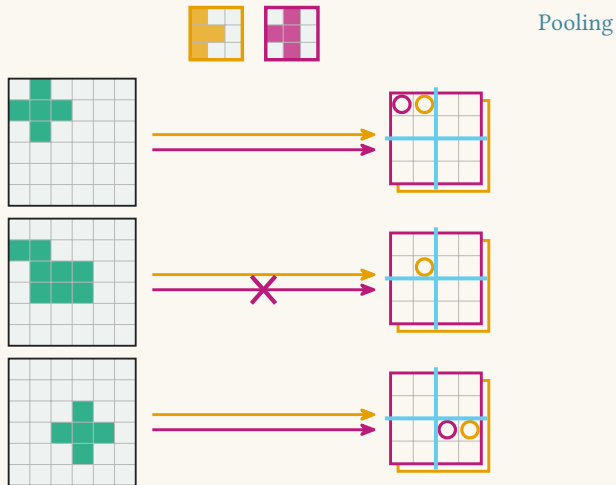
How to compute activations



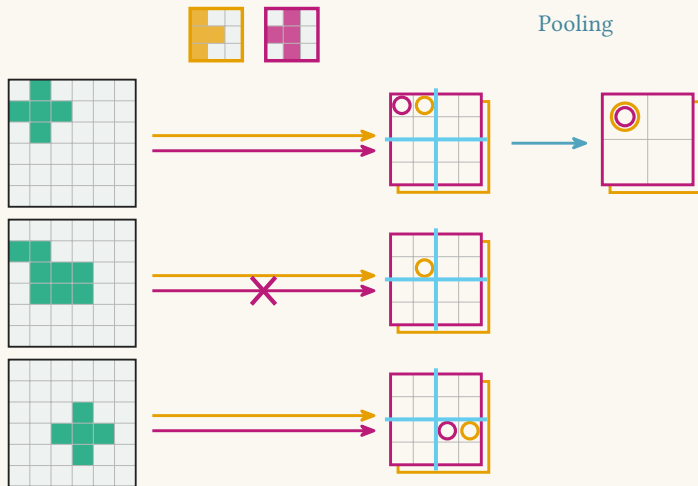
Recognizing islands that look like +, with pooling



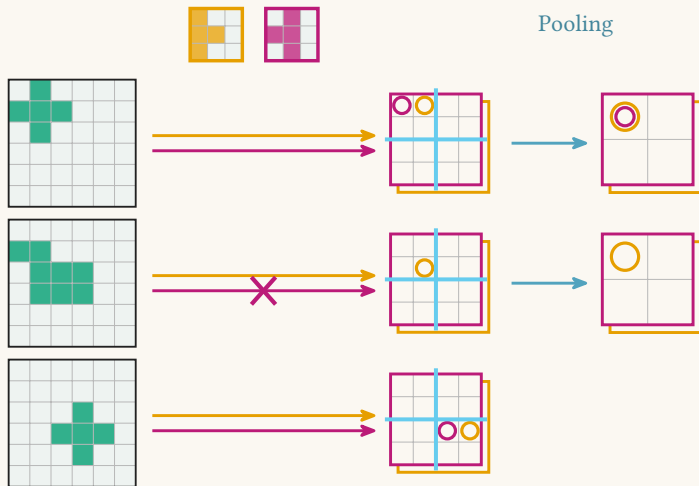
Recognizing islands that look like +, with pooling



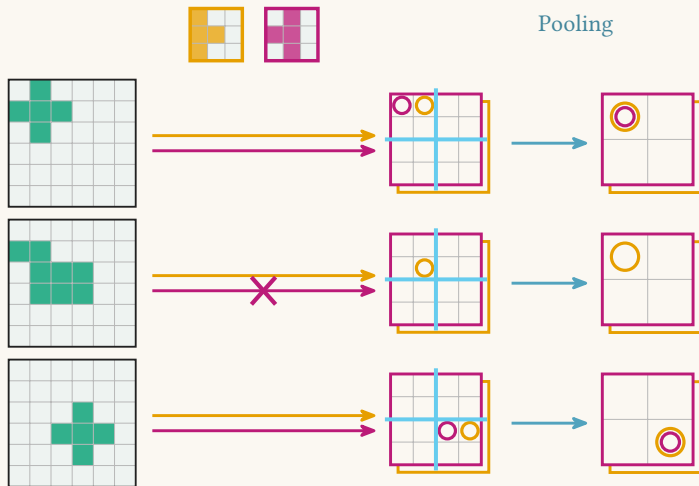
Recognizing islands that look like +, with pooling



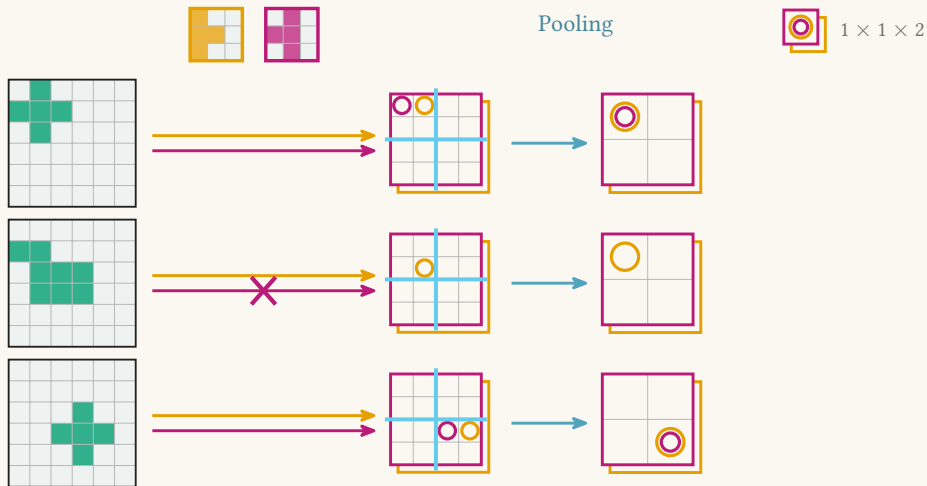
Recognizing islands that look like +, with pooling



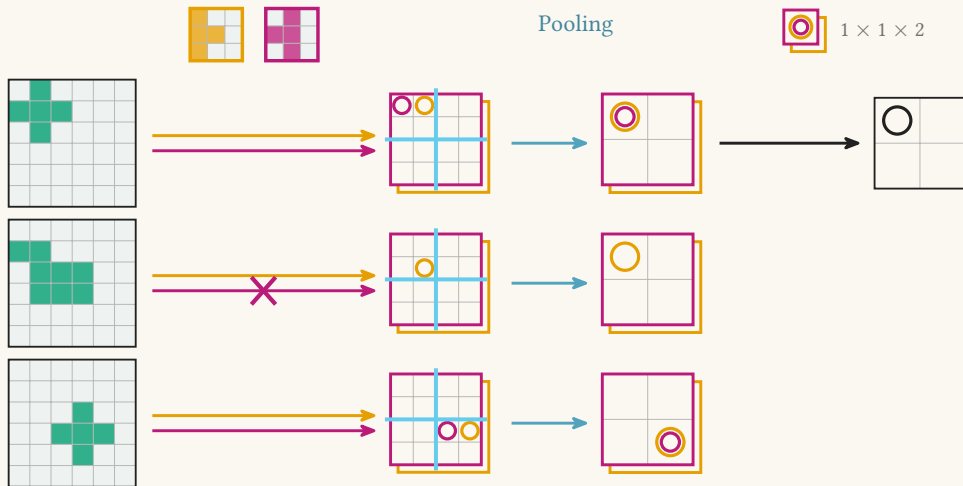
Recognizing islands that look like +, with pooling



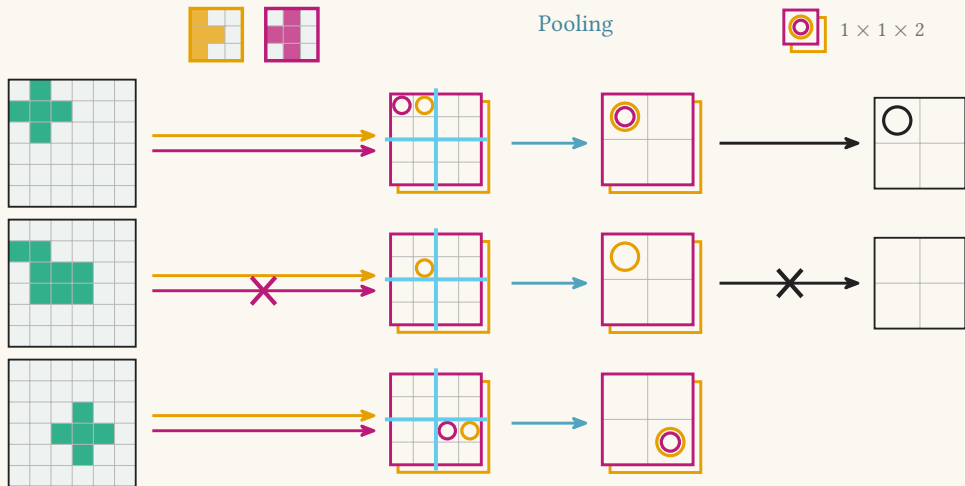
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